Examination of microwave properties

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We examine the focal length of a wax lens and the angular distribution of a microwave emitting Gunn diode. We then set up this source in combination with a metal mirror to use the resulting standing wave to determine the wavelength to $\lambda = (3.270 \pm 0.014)\,\mathrm{cm}$, whereas deploying a Michelson-interferometer yielded $\lambda = (3.280 \pm 0.028)\,\mathrm{cm}$. By sending collimated microwaves towards a $45\,^\circ - 90\,^\circ$ double paraffin prism we are able to show the inverse exponential dependence of the intensity of the evanescent wave for gap widths $\gtrsim 0.5\,\mathrm{cm}$ between the prisms. We are able to polarize microwaves by using a metal grid. Along a waveguide the polarization is maintained. Finally, by conducting the θ-rocking-curve method, we compute the lattice constant of a macroscopic crystal model to $d_0 = (4.39 \pm 0.22)\,\mathrm{cm}$.

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1 Introduction

The spectrum of electromagnetic radiation encompasses a broad interval of wavelengths, ranging from thousands of kilometers down to less than the radius of atomic nuclei. In this experiment we take a closer look at the lower end of its frequency spectrum. The wavelength of microwaves reaches from one meter to one millimeter and a plethora of phenomena and applications emerge from its properties. Hence the study of these is an integral part of the education of any physicist. For example they are widely used for heating in industrial processes [7]. Microwaves also play a big part in understanding our universe by examining the microwave background [8], which is said to be an leftover of the "Big Bang", the beginning of the universe. But being able to apply the properties of microwaves, it is needed a careful study of their behavior. In this experiment, we will do that by examining the angular distribution and the polarization properties of microwaves. It is also analyzed their wavelength and the formation of an evanescent wave. The determination of a lattice constant on a macroscopic cube by using the θ-rocking-curve method, will give an insight in the method that is used to analyze the interior of crystals via X-rays.

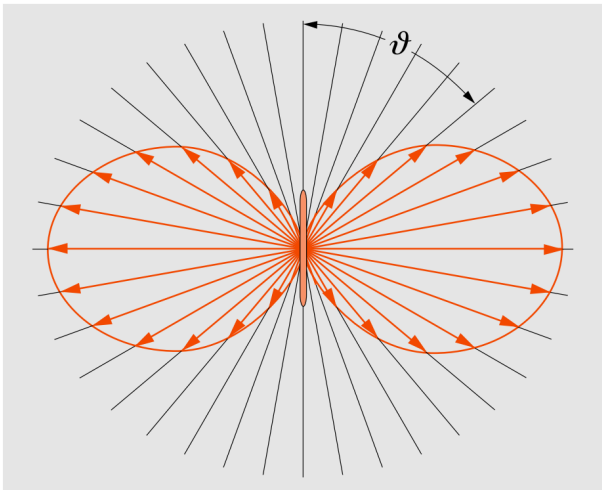

Figure 1: Symmetric angular radiation characteristic of a hertzian dipole. The longer an arrow is, the higher is the intensity of radiation at this angle. ϑ is the angle between the direction of radiation and the dipole axis. Orthogonal to the dipole axis radiation is the strongest. Only for $\vartheta = 90\,^\circ$ no radiation is emitted. [4]

2 Anglular distribution of microwaves

To examine the angular distribution of microwaves, we place the transmitter on a goniometertable, so

that the horn is located approximately at the pivot point. Then, after setting the millivoltmeter to its range of lowest resolution, we measure the angular distribution to both sides ±90° in matched steps to evaluate the drop-down of intensity. The measurements are done with a photo diode that is connected to an Agilent 34402A, thus our resulting voltage is a measure of microwave-intensity. To increase the sensitivity of our dectecor diode, it is applied an operating voltage of 84.8 mV inside. Even with no microwaves entering, our detector showed a voltage of −84.8 V. As a result all our measurements were systematically lowered by this operating voltage. We did correct his offset.

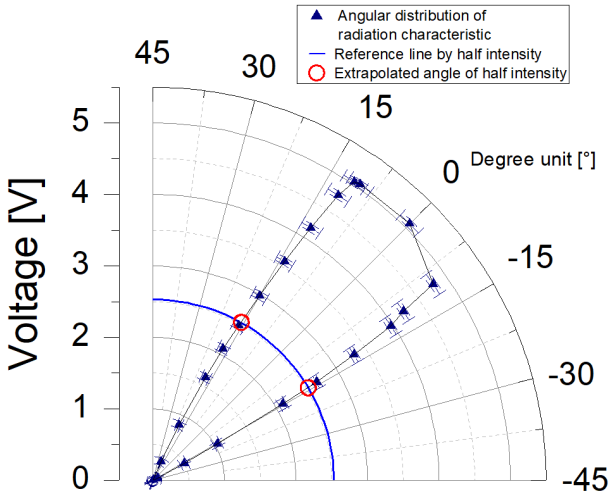

Figure 2: It is shown a polar plot of the measured voltage within the relevant angular range. The level of half intensity is marked with a blue line. Its crossing points with the voltage curve are highlighted red. The mean value of both is determined to be the half intensity angular: $\theta_{\text{half}} = (15.0 \pm 1.4)\,^\circ$. The angular uncertainty is chosen to be $2\,^\circ$.

In fig. 2 it can be seen that the angular range of the microwave radiation characteristic is about $\pm 16\,^\circ$ from the center. In order to reduce the background noise produced by reflected waves, we put the detector in close distance of about 39 cm in front of the microwave transmitter. Under these conditions, the extrapolated half intensity angular is evaluated to be

$$\theta_{\text{half}} = (15.0 \pm 1.4)\,^\circ$$

In comparison with the ideal radiation characteristic of a dipole, as shown in fig. 1, the measured angular distribution is more narrow and thus the microwaves are more focused. Also, it is apparent that the angular characteristic is not as symmetric, because the emitter is non ideal hertzian dipole. In the following measurements we considered that the irradiated object lies within the zone of greatest intensity; we assume the source to be point-shaped.

3 Focal length of a wax lens

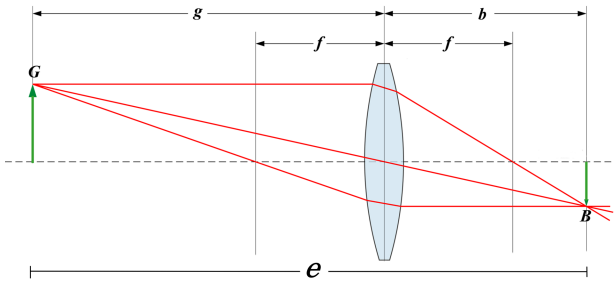

Figure 3: It is shown a construction of ray trajectories through a thin lens. The distance g between object G and lens, the distance b between lens and detector B, as well as the focal length f is marked. In our experiment, G is the microwave emitter.Adapted to [5].

To determine the focal length of a lens two pieces of information are needed: The distance between object and lens g, and the distance between lens and image b, as illustrated in fig. 3. The focal length f then reads:

$$f = \frac{bg}{b+g} \tag{1}$$

In the experiment we positioned the microwave transmitter on the optical bench. At a distance of $(g = 80.0 \pm 0.5)$ cm we located the wax lens. We estimated the error of reading to be 0.5 cm since the lens holder covered a small range of the benchmark, complicating the read-off. The reading error for the position of the emitter and the detector is chosen to be 0.2 cm. Then we measured the intensity of our microwaves after passing through the lens by successively lowering the distance between detector and lens in steps of 4 cm. After that we determined the maximum of the intensity to approximate the distance $b \approx 76.2$ cm and calculated the focal length, which was $f \approx 39$ cm. With this approximation the experiment is continued.

To determine the focal length more precisely, we plotted the detected voltage as a measure of intensity against the position of the detector, as can be seen in fig. 4. The error of voltage was estimated to 3 mV to account for interval of fluctuation on the digital display of the voltmeter.

The voltage curve does have a maximum, but it is overlaid with a periodic oscillation, which is characteristic for standing waves. This observation can be explained by considering that between detector

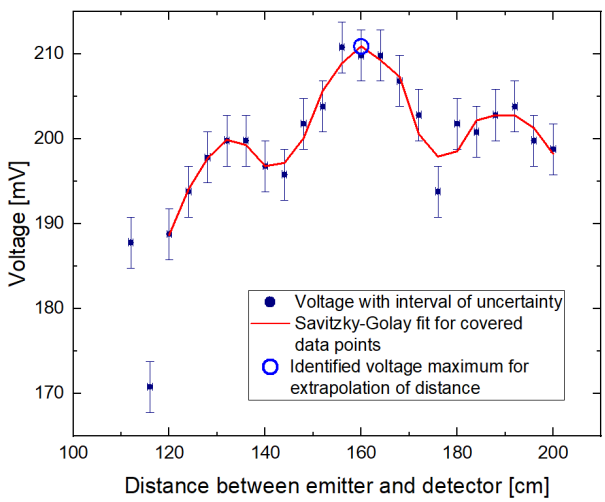

Figure 4: The measured voltage is shown applied to the distance e between microwave emitter and detector. The first two points were excluded due to strong data dispersion. The distance $e = (160.00 \pm 0.28)$ cm is extrapolated at the identified voltage maximum.

and lens, the microwave beam is partially reflected and thus forms a standing wave. To take account of the fluctuation that is caused by detection of microwaves that are reflected by our surroundings, we applied a Savitzky-Golay fit to our data with Origin. This method is performing a local polynomial regression and therefore it is well suited to remove noise without affecting the overall shape of the curve. In fig. 4 it is seen as red line. The first two data points are excluded due to strong data dispersion. With this fit we were able to identify the maximum of the voltage curve and extrapolate the distance e at which this maximum is reached. We arrived at $e = (160.00 \pm 0.28)$ cm. Thus, the focal length yields:

$$f = (40.05 \pm 0.19)\,\text{cm}$$

Next we placed the microwave transmitter in previously determined focal point of the wax lens to collimate the emitted waves, which is essential to perform the following measurements with precision.

4 Two methods for measuring the wavelength

If a wave is reflected by 180 °, it opposes and overlaps itself and thus forms a standing wave. Mathematically it is described with:

$$A(x,t) = 2A_0 \sin(2\pi\nu t)\cos(2\pi\frac{x}{\lambda}) \tag{2}$$

where A is the wave's amplitude, ν its frequency and λ is its wavelength [9]. It can be shown that the maxima and minima of this standing waves amplitude are spatially fixed and the distance between two neighboring nodes is an integer multiple of $\lambda/2$. Therefore, by knowing the exact distance b between n intensity minima of the standing wave, one is able to determine its wavelength by applying the relation:

$$b = n\frac{\lambda}{2} \tag{3}$$

where n is an integer. To examine the wavelength of the used microwaves, we placed the metal plate on the optical bench. One meter away from the lens, the collimated beam was reflected back on itself and as a result, a standing wave was formed between lens and metal. We then placed a diode into the beam and sampled 40 minima of intensity, recording the distance b traveled on the optical bench. The error of reading was chosen to be 0.2 cm. Using eq. (3) we were able to determine the wavelength to be:

$$\lambda = (3.270 \pm 0.014)\,\text{cm}$$

To validate our measurement, we then setup an Michelson-interferometer as shown in fig. 5.

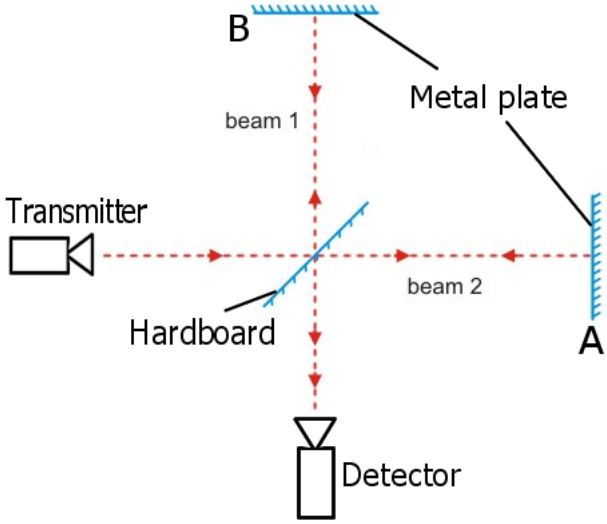

Figure 5: Setup for a Michelson-interferometer. The hardboard is used as a semipermeable mirror to split the beam of microwaves. The position of metal plate B is fixed whereas A is adjustable.

A hardboard served as a beam splitter, which splits the microwave beam into two rays. After they were reflected by the metal mirrors, the resulting beams reunite and travel into the detector. Since metal plate B is fixed and metal plate A can be adjusted, the distance which beam 2 travels is not constant and thus the relative phase of the two interfering beams depends the position of mirror A. If the phase difference between beam 1 and beam 2 is an integer

multiple of 2π, they will interfere constructive. If the phase difference is an odd integer multiple of π, the separate beams will cancel out. As a result, by lowering the distance b between hardboard and mirror A, we observed the detected voltage to vary periodically. We started at a maximum of interference. After the detected voltage passed though one minimum and reaches maximum again, we concluded that metal plate A was exactly moved by a distance of $\lambda/2$. For a higher precision we scanned 20 maxima of intensity and noted the distance b between the starting- and ending position of mirror A. Then we applied eq. (3) and determined the wavelength to

$$\lambda = (3.280 \pm 0.028)\,\text{cm}.$$

For the error of reading we chose again 0.2cm for each read-off. According to its producer the frequency of the source is $f = 9.35\,\text{GHz}$. Using the relation $\lambda = c_0/f$, where c_0 is the speed of light in vacuum, this frequency corresponds to an expected wavelength of $\lambda \approx 3.206\,\text{cm}$.

Comparing both methods, one may object the error for the interferometer method is higher than assumed: The beam splitter was adjusted by hand, meaning that the angle with the incident wave was different from the assumed $45\,^\circ$. This results in an non-constant offset for the position of antinodes, because the way the wave traveled forms an acute triangle. This difficulty does not occur for the first method, yet since both values agree within the margin of error, we calculate the wavelength by averaging the separately obtained results, hence:

$$\lambda = (3.275 \pm 0.072)\,\text{cm}$$

This value definitely lies within the wavelength range of microwaves. Within their interval of uncertainty, the determined wavelength in both of the experimental setups do not agree with this expected value. Nonetheless, within the error interval, both methods of measurement reproduced the wavelength value and thus the deviation from the expected wavelength seems to be system-immanent. At the time of our measurement, the emitter was noticeably heated which may be a source of error due to the temperature-sensitive spectrum of Gunn-diodes. Also, it has to be added that the emitted frequency might have changed due to wear-down.

5 Tunnel effect for microwaves

In quantum mechanics, the wave function of a particle in a finite potential well has a non-zero amplitude inside of and behind the (sufficiently thin) barrier, meaning that by a small probability the particle can overcome the potential trap and thus leave the well, although in a classical sense its energy does not suffice. The same is true for an electromagnetic wave that is being total internal reflected, because the wave partly enters the phase with a lower refractive index. Maxwell's laws require the intensity of the transmitted wave to decrease exponentially with the distance inside the other medium with a lower refractive index. If another medium of higher refractive index than the filling of the gap is brought close enough, the transmitted (evanescent) wave couples on it, and energy is thus transmitted across the gap (frustrated TIR, *FTIR*).

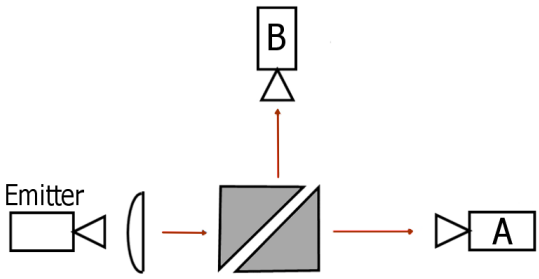

Figure 6: Setup for measuring the evanescent (detector A) and reflected (detector B) wave. The emitter sends out microwaves that get collimated by a lens and enter the double prism (grey). Adapted from [1].

In this experiment we examined the intensity of the reflected and transmitted microwave entering a double $45\,^\circ - 90\,^\circ$ prism. The set up can be seen in fig. 6. By increasing the distance d of the prisms, the intensity of the passing and reflected wave was observed.

For the detected signal in position A, we expect an exponentially decreasing intensity, whereas detector B should measure an exponential increase in intensity. This is because the closer the prisms are, the more photons tunnel while less photons get reflected correspondingly.
The obtained results are shown in fig. 7 . We used a paraffin double prism. The voltage error was estimated to 3 mV due to fluctuations on the display of the voltmeter, the error for the distance between the prisms to 0.5 mm.

In fig. 7 (a) the expected behavior can be observed only for $d \gtrsim 0.5\,\text{cm}$, where the data appears to depend linearly on the distance. For lesser d, this trend vanishes. This can be explained by reflections inside of the second prisms at the junction prism-air, resulting in less intensity observed than expected. The reflections cause the deviation from the exponential curve for small distances d since the intensity I is inversely proportional to the square of the distance: $I \propto 1/d^2$. The result in fig. 7 (b) is unexpected: The

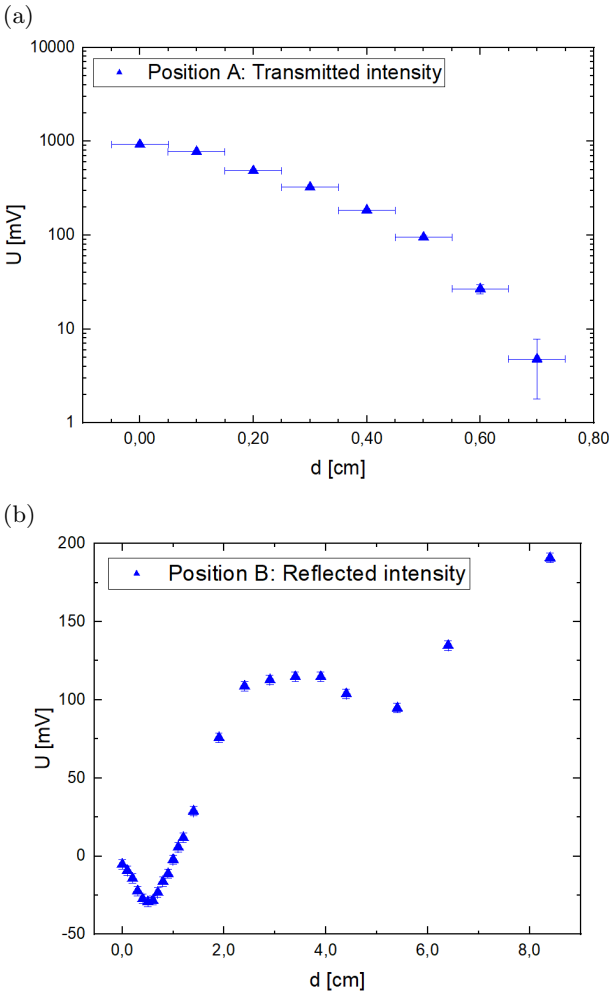

Figure 7: (a) Measured intensity of the evanescent microwave for two paraffin prisms at a distance of d; for $d = 0\,\mathrm{mm}$ the prisms were sandwiched back-to-back. The voltage axis has been logarithmized.
(b) Measured intensity of the reflected wave for the same setup as in (a). Linear U-axis.

intensity does increase with distance but not purely exponentially, though. There seems to be another function overlaying the exponential. We surmise the development of a standing wave in the gap between the two prisms causing the oscillations on top of the exponential increase. Our result is comparable to the data obtained by [2].

6 Polarization of microwaves

To examine the polarization properties of microwaves we placed the emitter on the optical bench. behind the lens which collimated our beam, we put a metal grid between lens and detector. Since the gaps of the metal rods was in the scale of the emitted wavelength, the grid worked as a polarization filter. In fig. 8 the effect of a polarization filter is depicted. If the electric field vector of the incident wave is parallel to the metal rods, it will induce an alternating current within them. According to Huygens' principle, this alternating current then radiates waves in all directions, which have opposite phase with respect to the incident wave.

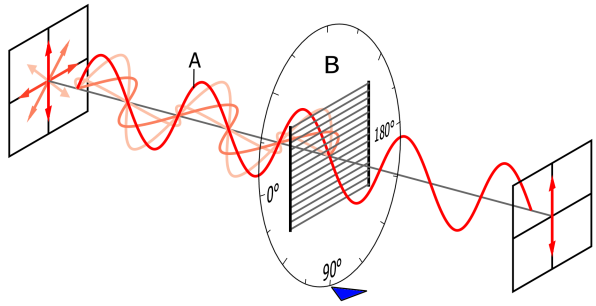

Figure 8: Schematic construction of the experiment. The electric field of a non polarized electromagnetic wave traveling through a polarization filter (B) is shown by bold orange. Afterwards it is vertically polarized, since the electric field oscillation parallel to the metal rods are blocked off. In our setup the source only emits a vertical polarized wave (A). The reading mark is shown as blue triangle. Adapted from [6].

As a result, only the electric field oscillations orthogonal to the metal rods will pass the filter unhindered, because they do not induce any alternating current.
For different positions of the metal grid we measured the voltage which is representative for the microwave intensity. We logged data points from $0\,^\circ$ to $180\,^\circ$ in steps of $10\,^\circ$ and estimated the error of reading to be $2\,^\circ$. The collected data was processed and is depicted in fig. 9.

Malus law provides the theoretical description of intensity depending on the angle of a polarization filter. For our setup it has the form

$$I(\theta) = I_0 \sin^2(\theta), \tag{4}$$

yet formula does not explain the shape of the measured intensity curve alone, as can be seen in fig. 9, though. After removing the metal grid and turning the detector diode on its side by a rotation of $90\,^\circ$, we observed no signal anymore. We therefore conclude that the detector itself behaves like a polarization filter. Applying our conjecture to the theoretical considerations, what we actually see in fig. 8 is the intensity distribution of two polarization filters connected in series. It is thus described with:

$$I(\theta) = I_0 \sin^4(\theta) \tag{5}$$

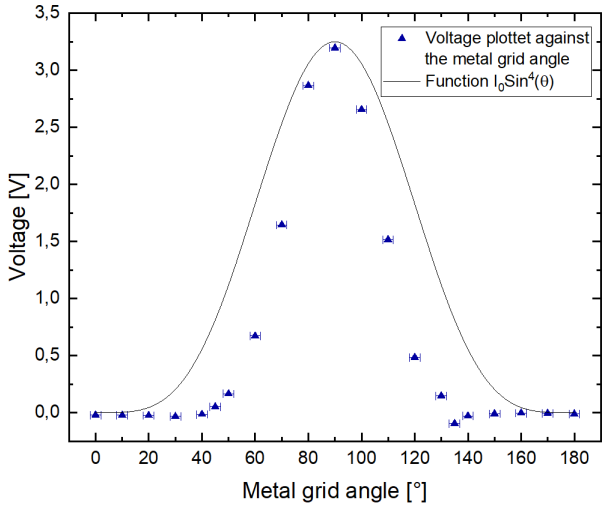

Figure 9: Plot of the measured voltage against the angle of the metal grid, starting from 0 ° to 180 ° in steps of 10 °. Extra data points are taken at 45 ° and 135 °. The maximum of intensity is clearly perceivable at 90 °. Fig. 8 gives a snapshot of this position. It is applied an example function for $I_0 \sin^4(\theta)$ inside the graphic with $I_0 \approx 3.25\,\mathrm{V}$.

We applied a fit of this function to our data. It is apparent that the measured intensity curve is systematically narrower than theoretically expected. This deviation might be caused by the fact that the used metal grid is non ideal polarization filter since it also reflects the microwaves in various directions. The same argument accounts for the detector we were using, that is made of a photo diode, which only observably behaves as a polarization filter, but provides no guarantee that it is described with Malus law given in eq. (4).

7 Determination of the lattice constant of a cubic crystal

In solid state physics, the θ-rocking-curve method uses x-ray to explore the characteristics of an atomic lattice such as the distance between each layer, called the lattice constant, that usually has the value of a few Ångström ($1\,\text{Å} = 10^{-10}\,\mathrm{m}$) for real crystals like NaCl. In this experiment, we used this technique to determine the lattice constant of a macroscopic crystal in the shape of a styrofoam cube doted with aluminum wrapper balls such that a cubic lattice emerges. Bragg's law states that monochromatic radiation of wavelength λ interferes constructive under the angle θ if

$$n\lambda = 2d_{hkl} \sin\theta \tag{6}$$

is true when being diffracted from a lattice with a lattice constant of $d_{hkl} = d_0/\sqrt{h^2+k^2+l^2}$ in the crystal direction $[hkl]$, where d_0 is the distance between two next neighboring aluminum balls. n denotes the order of the maximum.
As a consequence of eq. (6), one needs at least radiation of the wavelength $\lambda = 2d_{hkl}$ in order to examine a lattice. Thus for real crystals, x-rays of wavelengths between 10 nm and 10 pm are necessary, yet since the crystal we used has a lattice constant of a few centimeters, microwaves are sufficient.
To determine the lattice constant we chose a setup as shown in fig. 10.

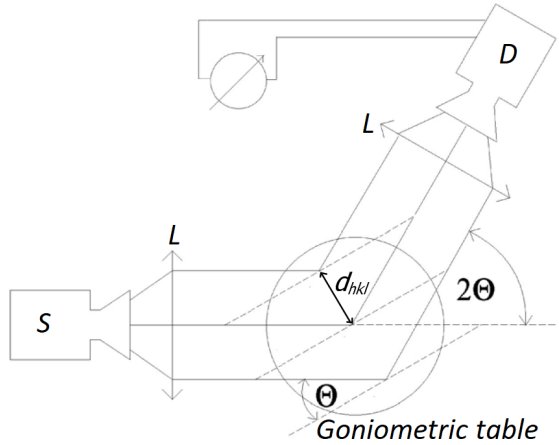

Figure 10: Setup to determine the lattice constant of a sample on the goniometric table. Microwaves emitted by the source S are collimated by the lens L, get diffracted and are focused by the second lens L to be recorded by the detector D. Adapted from [1]

The wavelength of the source was already determined to be $\lambda = (3.275 \pm 0.072)\,\mathrm{cm}$ in chapter 4. For the crystal direction [100] we placed the styrofoam cube accordingly on the goniometric table and recorded the voltage of the receiver depending on the angle θ between the sample and the incident beam. In this procedure a rotation of the table by θ means the detector had to be rotated by twice the angle. The results are shown in fig. 11 where we added a line of reference to distinguish between the first maximum and the background noise as well as a bandwidth. The latter serves as an estimate of the maximum error of the reading point, which is the distance of this point to the farther edge of the bandwidth. The former is the average voltage of the red dots in fig. 11 that are not within the interval indicated by the dashed lines. The results for read off values (represented by the red crosses in fig. 11) and corresponding lattice constants are listed in tab. 1.

Qualitatively, these values are in the right order of magnitude but there are several problems: From

n	$\theta/°$	d/cm
1	23.0 ± 2.0	$4.19 \pm 0,36$
2	40.0 ± 2.0	5.09 ± 0.24

Table 1: Angles θ under which the n-th maximum was observed and resulting lattice constant d obtained for the θ-rocking-curve method with microwaves diffracted by a macroscopic cubic crystal model.

eq. (6) for the obtained lattice constant one would expect to find the maximum of second order in the interval between $40\,°$ and $50\,°$, yet this can not clearly be observed in fig. 11. We assume the intensity of the signal there to be too weak to be observed compared to the background noise. In theory, one would expect zero intensity for every angle except for $\theta \approx 0°$ (total reflection) as well as in a small region around the maxima. Also, the values for the lattice constant are not congruent within their margin of error.

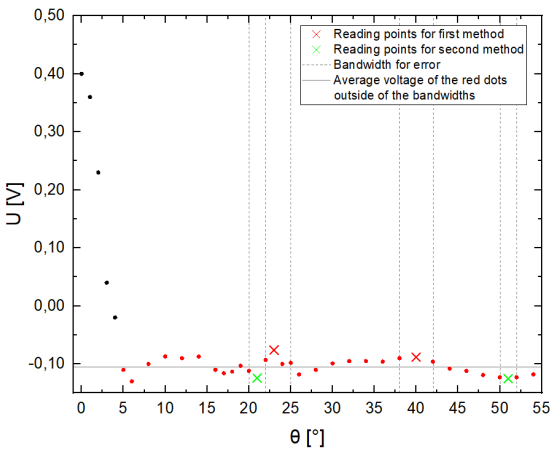

Figure 11: Intensity after diffraction of microwaves incident on the 100-plane of a macroscopic crystal model (styrofoam cube doted with aluminum balls to resemble a cubic lattice). The angle θ is measured between the plane of the sample and the incident wave.

In chapter 2, where we discussed the angular distribution of the microwave emitter, we found out that it can not be approximated by a point source. Consequently, not all microwaves are sent towards the first lens in fig. 10, some are sent into the space leading to a constant background radiation field in the laboratory. Hence another way to look at the data shown in fig. 11 is by assuming the peaks interfered destructively with this background field of radiation. Interpreted this way, we have to read off the local minima instead of the local maxima, as it was done

n	$\theta/°$	d/cm
1	21.0 ± 2.0	4.57 ± 0.43
2	51.0 ± 1.0	4.21 ± 0.11

Table 2: Angles θ under which the n-th minimum was observed and resulting lattice constant d obtained for the θ-rocking-curve method with microwaves diffracted by a macroscopic cubic crystal model.

before.

The new results are listed in tab. 2. The angles under which a minimum was found are represented by green crosses in fig. 11, where we added the cross at the second minimum afterwards to clarify we read off right between the adjacent data points that carry the same voltage value. The minimum observed at $\theta = 6\,°$ can be ignored, being an artifact of underlying *sinc*-function behavior for $\theta \approx 0\,°$.
For this method, the lattice constant agree within the margin of error. Given that the reading precision is the same, technically, the higher the order of the maximum, the more precise is the value for the lattice constant since the relative reading error $\sigma\theta_\mathrm{n}/\theta_\mathrm{n}$ is smaller. In our case, the reading precision for both peaks is not the same. We thus refrain from calculating a weighted average, but instead determine the lattice constant as the unweighted average of the values given in tab. 2:

$$d_0 = (4.39 \pm 0.22)\,\mathrm{cm}$$

It has to be added though that the data we obtained complicated the evaluation because of two reasons: First, we do not know whether to use maxima or minima and second, either were not clearly identifiable. Thus the above value for d_0 might carry a larger error.

8 Properties of a circular waveguide

Waveguides are resonators with two open ends, allowing waves to propagate on its interior from one end to the other with minimal loss of energy by restricting the transmission of energy to only the direction of the guide. For microwaves, a hollow conductive metal pipe is usually the choice.
The aim of this section is to show that the used metal waveguide has excellent transportation properties for microwaves, despite its ribbed shape and length. For this we placed the horn of the same microwave source that has been used throughout this paper directly in front of one end and positioned a receiver directly behind the other end to detect the remaining intensity

of microwaves. To make sure this signal was transmitted through the pipe and not just background noise, we inserted a metal mirror between the emitter and the waveguide to shield off incident microwaves. The signal intensity instantly dropped to the level of background noise observed in chapter 2. We also varied the distance between source and tube and observed a decrease in intensity for an increase of distance. Thus we conclude the waveguide is able to transmit microwaves. Finally, we examined the polarization of the transmitted waves by placing a polarization filter between the receiver and close end of the hose. The frequency depended heavily on the angle of the filter relative to tube's rim. For some angle, the intensity almost reached the level of the source with no tube in between, perpendicular to that the observed intensity dropped to the level of background noise. This means the waveguide preserves polarization.

9 Conclusion

We measured the angular distribution of the microwave emitter and evaluated its radiation characteristic to be similar to dipole radiation. Next we determined the focal length of a wax lens, which we then used as a collimator for the microwave beam. We arrived at $f = (40.05 \pm 0.19)\,\mathrm{cm}$. Further by letting a standing wave arise between mirror and lens, the wavelength of the microwaves was evaluated to be $\lambda = (3.270 \pm 0.014)\,\mathrm{cm}$. Then we performed a Michelson Morley experiment. With this method the wavelength was measured to be $\lambda = (3.280 \pm 0.028)\,\mathrm{cm}$ and thus it was consistent with the first measurement. Furthermore the tunnel effect for microwaves was examined. We were able to show, that an evanescent wave was formed, which can be described by an exponential law. We then observed the polarizing properties of a metal grid. The resulting intensity distribution was better described with a $\sin^4(\theta)$-function. By turning the detector $90\,^\circ$ without any metal grid, we could evaluate the conjecture that the detector itself behaves like a polarization filter. Finally we determined the lattice constant of a cubic crystal by measuring an θ-rocking-curve. We determined it to $d_0 = (4.39 \pm 0.22)\,\mathrm{cm}$.

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